Pi Extension Module

— Review and Debugging Notes —

Fides Thérèse

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1 Purpose and project position

The extension adapts the existing Infinite Generation Two prompt and metadata to Pi without emulating DeepSeek Harness. Pi loads the default factory in `index.ts` from the package manifest. This module is a leaf customization inside Pi, not a provider, standalone agent, or browser plugin. The module's architecture diagram is in the specification and implementation contracts are in `docs/DTD.tex`.

2 Inputs, outputs, and relationships

The module consumes Pi lifecycle events, slash-command arguments, the startup off flag, and completed assistant content. Its only asset is the original `prompts/infinite-gen-2.md`. Outputs are an owned structured system-prompt section, a read-only profile-tool result, and an English keyed UI status. Headless commands report a non-triggering custom message. Application code does not write files, open network connections, or spawn background work.

The adapter depends on two pure-facing helpers. The profile helper reads and hashes the asset and constructs TypeBox tool schemas. The status helper extracts text, scans inherited bilingual phrases, and formats labels. Neither helper depends on the adapter. Pi provides the host API and TypeBox; Node's filesystem and crypto standard libraries are the only production I/O libraries. No separate provider SDK is required.

3 Control and state flow

The factory verifies the prompt before registering any capabilities. Before each run, the adapter assigns or deletes only its `infinite_gen2` prompt section. The complete prompt is never replaced. The parameterless `infinite_gen2_profile` tool returns the unchanged prompt with upstream metadata as text, details, and structured JSON.

Session start resets state from `--infinite-gen2-off`. Commands toggle injection and monitoring only for that session without clearing active-run state. Run start clears old results and marks the status busy. Completed assistant messages update the last text verdict; thinking, tool calls, user messages, and empty content are

ignored. Provider errors and cancellations, including automatic compaction failures, have separate outcomes. Final settlement ends the busy indicator after Pi's retries and queued work, and shutdown clears the UI key. Tree navigation clears stale results without changing the toggle. No state is reconstructed from abandoned branches or persisted.

4 Key decisions and limits

The prompt is an immutable exact-byte asset. An invalid fingerprint or missing file stops extension loading; no fallback prompt is supplied. The original prompt version remains 0.2.3, independently of extension version 0.3.0. The status detector retains the upstream phrases and exclusion window. Its verdict describes wording only: quotes can match, and a pass does not prove model compliance or prompt effectiveness. It never modifies output or retries a model request.

Pi owns model access, credentials, tool approvals, and operating-system permissions. A later full-prompt replacement by another extension can override this section. Disabling cannot retract instructions already sent or prompt content returned by the profile tool. New or reloaded sessions reset toggles; persistent preferences and browser animations are deliberately absent.

5 Relevant verification

Run `npm test` for actual Pi SDK loader, event-runner, command, tool, state, and integrity checks; `npm run check` for strict TypeScript; `npm run smoke` for isolated offline CLI loading; and `npm pack --dry-run` for distribution contents. Tests isolate the agent directory, use in-memory sessions, dispose resources, and make no model calls. Live DeepSeek response behavior is outside the verified boundary.